Reliable data, with traceability

Project overview

Prepare a large historical retail workbook for transparent analysis while preserving its source records. Mixed identifiers, missing customer details, repeated rows and signed transaction values require explicit treatment before business users can interpret totals.

1,067,371
source rows preserved

1,041,670
gross positive sales lines

7 / 0 / 0
tests: pass / fail / skip

Dataset and scope

UCI Online Retail II contains transaction lines from a UK-based online retailer, covering December 2009 to December 2011 across two worksheets. The eight original fields describe invoice and product identifiers, description, quantity, invoice date, price, customer ID and country. The historical workbook remains the authoritative source.

Reconciled output counts

Worksheet	Source = cleaned	Sales subset
2009-2010	525,461	511,565
2010-2011	541,910	530,105
TOTAL	1,067,371	1,041,670

Two delivery views

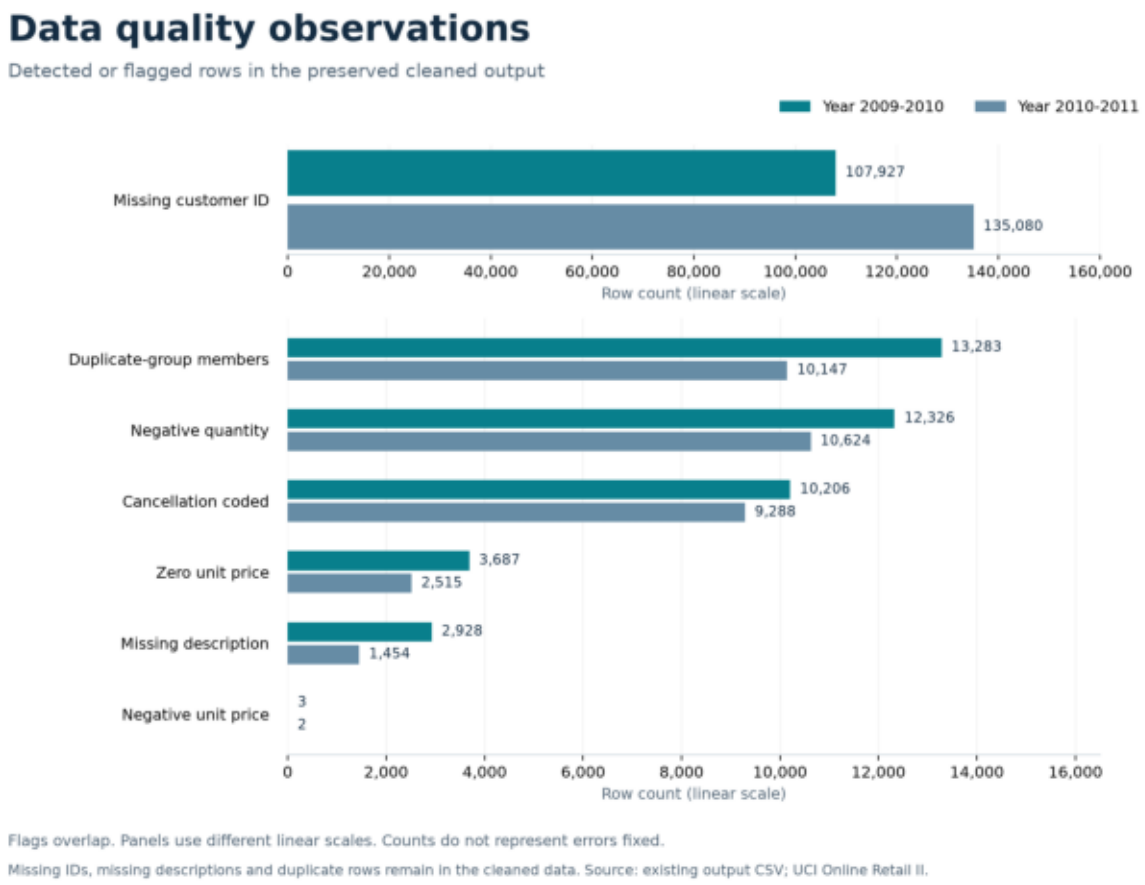
The cleaned CSV retains all 1,067,371 lines, original source values, worksheet origin and source row references. The sales CSV contains only positive-quantity, positive-price, non-cancellation lines. The other 25,701 records are excluded only from that sales view; they remain in the cleaned output.

Source attribution

Chen, D. (2012). Online Retail II [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5CG6D. CC BY 4.0; standardized and sampled. Self-directed portfolio project, not a paid client engagement.

Observed quality and treatment

Detected or flagged records are review observations, not necessarily errors. Categories overlap. The chart compares worksheets rather than implying that business events were corrected or removed.



Measured before / after preservation

Measure	Before	After cleaning
Source row count	1,067,371	1,067,371
Missing customer IDs	243,007	243,007
Missing descriptions	4,382	4,382
Duplicate occurrences*	12,133	Retained; no removal

*Duplicate occurrences count rows after the first in each repeated group (initial assessment). The duplicate flag marks all 23,430 group members. No missing IDs or descriptions were filled; prices were not imputed.

Validation and delivery guidance

Cleaning and standardization

The existing Pandas pipeline processes worksheets sequentially, standardizes headers and surrounding whitespace, parses numeric and date fields, and keeps invoice and stock identifiers textual. Original values remain in source audit columns; missing customer IDs and descriptions remain missing. Row-level quality flags support review without silent deletion, filling or deduplication.

Recorded validation evidence

The final verification report records 7 pytest tests passed, 0 failed and 0 skipped using the existing virtual environment. It independently verifies full CSV readability, exact row counts, unique populated source row IDs and complete worksheet row coverage. The workbook's recorded size and modification time were unchanged. These are technical checks; no commercial performance improvement has been measured.

Classification and signed amounts

Status precedence is: review for missing required values, zero quantity or negative price; then cancellation-coded invoice; then other negative quantity; then zero price; otherwise normal sale. Independent flags preserve overlapping conditions. line_total is quantity multiplied by unit price when both are finite numbers; its sign follows the operands.

Business assumptions and limitations

Negative quantities and C-coded invoices may represent returns, corrections or other business events; the data alone does not verify refunds. Duplicate records remain in both outputs when eligible. The sales subset is gross positive sales lines, not verified net revenue. No invoice-level netting or price imputation is performed. Currency calculations use floating-point values without a currency-rounding policy.

Handoff and correct import

Use the delivery guide and all-column data dictionary for interpretation. The sample workbook contains 805 deterministic, stratified rows with explicit Excel text identifiers. It overrepresents rare statuses and must not be used for population totals. CSV has no type metadata: import invoice numbers, stock codes and customer IDs as strings; parse dates explicitly. Full CSVs stay outside the portfolio folder and are ignored by Git.

Reproduction and next decision

Run src/build_portfolio.py to regenerate this package from the existing outputs and verification reports. The cleaning pipeline is not rerun by the generator. Business owners should agree on refund interpretation, duplicate handling and accounting precision before using these files for financial reporting. Nothing has been published or uploaded.